

## Information Systems

### Web Technologies and Service-oriented Architectures for Organizations

#### IDENTIFICATION

CODE : IF-4-WASO  
ECTS : 1.0

#### HOURS

Lectures : 6.0 h  
Seminars : 0.0 h  
Laboratory : 8.0 h  
Project : 0.0 h  
Teacher-student  
contact : 14.0 h  
Personal work : 10.0 h  
Total : 24.0 h

#### ASSESSMENT METHOD

#### TEACHING AIDS

#### TEACHING LANGUAGE

French

#### CONTACT

M. GRIPAY Yann  
[yann.gripay@insa-lyon.fr](mailto:yann.gripay@insa-lyon.fr)  
MME SERVIGNE Sylvie  
[sylvie.servigne@insa-lyon.fr](mailto:sylvie.servigne@insa-lyon.fr)  
Phone : 0472438483

#### AIMS

The first objective of this module is to study a methodology for designing service and micro-service -oriented architectures for IS applications, based on the UML formalism: a) analysis of functional situations; b) identification of business objects; c) identification of services and service layers; d) specification of services; e) distribution on the technical architecture.

The second objective is the discovery of technologies for the implementation of these architectures: Web services, application servers, data management middleware, communication middleware.

The target skills in this module are:

- 1) Apply a method for analysis and design of IS:
  - Using UML models as part of an IS design method
  - Analyze business and requirements of clients
  - Identify business objects, business services, and associated GUI
- 2) Design application cartography and technical cartography:
  - Identify application blocks and information flows
  - Design a technical architecture to deploy application components
- 3) Design and implement applications for IS:
  - Write IS specifications documents
  - Implement an application from IS specification documents

#### CONTENT

- Methodology for design of service-oriented Information Systems and development of multi-tier application architectures
- Technical Architectures for Web Services
- Technologies of Application Servers
- Middlewares for data management and communication

#### BIBLIOGRAPHY

Micro Service: Patterns and Applications, L. Krause, 2015

#### PRE-REQUISITE

Application Development for Information Systems  
UML Modeling  
Databases  
Algorithmic and Object-oriented Programming  
Java development